

OPENING LECTURE

Time Is Money, Yet Finance Has No Time!

So Why Do the Models Forget to Check the Clock?

EC3108 Financial Economics Dynamics - Intro lecture

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Road map

Part 1 The crime scene: four exhibits, all missing the same thing

Part 2 What “having time” would even mean

Part 3 What happens when you actually add it — two plot twists

Part 4 Even the clock you added might be lying to you

Part 5 The whole semester, in one sentence, and questions

Fair warning. This lecture is a little cheeky about a discipline you've invested in and may well be building a career within. That's on purpose. It nudges you to step back from familiar frameworks and see the field using a critical lens - you should never fully trust a model you haven't first been allowed to laugh at! 😊

“Remember that time is money.”

— Benjamin Franklin, *Advice to a Young Tradesman*, 1748

Every finance textbook gestures at some version of this sentence, usually on page one.

It is, by a wide margin, the field's most quoted line — and arguably its founding slogan.

Today's question: if it's true, how much time is actually *in* the models you have learnt before?

Let's actually check.

*Four exhibits. All standard, all correct, all things
you either already know.
See if you can spot what's missing before I say it.*

Part 1 The Crime Scene

Four exhibits from the standard finance curriculum. All four are evaluated at a single instant.

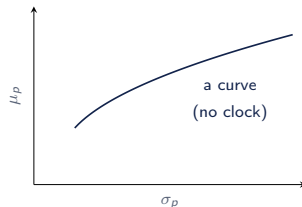
Exhibit A — The Efficient Frontier

Minimise portfolio variance for a given target mean and trace the resulting curve in (σ, μ) space.

It is a beautiful, genuinely useful object.

Question: when does this happen?

Answer: There is no “when.” It is computed once, from a covariance matrix that is simply handed to you.



A frontier is not a trajectory. It's a floor plan - a photograph of a room, not a video of someone moving through it.

Exhibit B — The CAPM

$$\mu_i - r_f = \beta_i (\mu_M - r_f)$$

β_i is a number.

Yes, you *estimate* it using historical data over time — but the equation it plugs into is a cross-section: a relation among assets *at one moment*, not a path for any one of them.

Nothing in the equation says what β_i does next week.

It's a class photo. It tells you everyone's relative height on picture day. It does not tell you who is still growing.

$$P = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t}$$

“Wait — there’s a t right there. Surely *this* one has time in it?”

It has a t the way a photograph of a marathon has runners in it: present, motionless, mid-stride, forever.

The sum runs over future dates, but the formula itself never moves. It is computed once, collapsing all future values back to the present — **the inverse of the Big Bang**, a whole timeline imploding into a single number. Nothing updates when new information arrives.

A subscript is not a clock. D_t just means “a number, indexed by a letter that reminds you dates exist.”

“All available information is already reflected in the price.”

Read that sentence again. It is a claim about *right now*.

It says nothing about how the price got here, and very little about where it goes next, except “you can’t tell from here.”

A snapshot claim, dressed up in the vocabulary of information.

Every one of these four results is genuinely useful. None of them is wrong. You will learn, use, and rely on all four.

But look at what they have in common: each is evaluated at a **single moment**, holding a fixed set of numbers, with nothing inside the model that moves on its own.

None of them has a working clock inside it.

TIME IS MONEY

*- for a field built on that slogan, remarkably few of
its workhorse models have a clock in them!*

It's a bit like a restaurant called *Fresh Catch Daily* that serves you a can of tuna and reminds you the label says "packed on a boat."

To Be Fair to Franklin, and to Fisher

Irving Fisher's 1930 separation theorem genuinely does have two dates in it: today and tomorrow. Arguably finance's first serious attempt at time.

But watch what happens to it: it collapses into one decision (how much to invest) with no calendar in it at all, and one decision (consume now or later) that only ever compares *two* moments, never three, never a whole horizon.

Preview. Block 1 of this course is essentially Fisher's two-date world, stretched out and formalised by Tobin, Markowitz, the CAPM but never actually handed a third date, let alone an infinite one.

Part 2 What Would “Having Time” Even Mean?

Before we add it, let's review what's currently missing.

A Photo vs. a Video of a Pendulum

the picture on this slide tells you:

Where the pendulum is, right now.

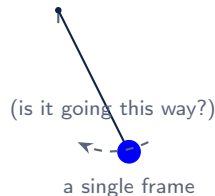
what the photo does *not* tell you:

Which way it's moving

How fast

Whether it's slowing down

Whether it's about to stop



This is exactly the difference between a static model's output (a point, a curve, a cross-section) and a dynamic model's output: **a path**, with a before and an after.

Three Ways Time Actually Enters a Model

A state variable that carries history forward. Today's wealth remembers yesterday's decisions; it is not handed to you fresh each morning.

Expectations over what hasn't happened yet. Not just what you know, but what you're guessing and how confident you are in the guess.

Feedback. Today's decision changes tomorrow's menu of choices, not just tomorrow's bank balance.

If none of these three things are present, writing a subscript t on a variable doesn't make a model dynamic. It just makes a static model wear a wristwatch.

Part 3 What Happens When You Actually Add It

Two plot twists. You will not guess the second one!

Same Question: Plot Twist #1 Samuelson (1969)

Take the identical portfolio-choice question from Block 1.

But now: many periods. Invest every period. The investor *knows* she'll be back next period to choose again.

Surely this is now dramatically more complicated?

Let's see.

Sometimes Time Changes... Nothing

Samuelson's myopia theorem: under CRRA utility and i.i.d. returns, the many-period answer is *identical*, at every single date, to the static one-period answer.

The static formula, it turns out, was accidentally right all along, but only in this specific instance single period action repeated over time

This is why repeating the same **static problem** in Block 1 over time is not dynamic

What if the investment opportunities themselves change?

What if expected returns drift, a little predictably, over time?

Plot Twist #2 — Now Time Changes Everything

Merton introduces a brand-new term in the optimal portfolio: the **intertemporal hedging demand**.

The investor now wants to hold assets that do well *precisely when the future gets worse* - sort of **insurance** against a moving target, bought before you know you'll need it.

No static model in Block 1 has a slot for this. It is as if the new term is invisible from a single snapshot models - uncovered by Merton

So Which Is It - Decoy, or Revelation?

Adding time to a model is like watching Part 2 of a story.

Sometimes Part 2 is exactly like Part 1, just longer *(myopia)*

Sometimes Part 2 reveals the first movie was hiding something the whole time.
(hedging demand)

You do not get to know which one you're in without actually doing the dynamics -
This is the **Mainstream dynamic models**, covered in Block 2 of this course.

Part 4 Even the Clock Might Be an Illusion



Block 2 fixed one blind spot. It quietly kept another

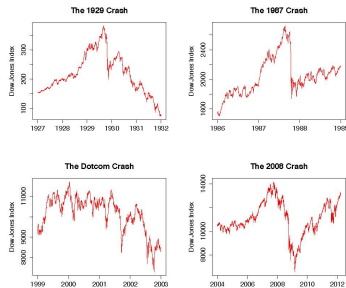
What Merton's Investor Still Secretly Knows

Merton's investor worries about a moving opportunity set - genuine progress over Block 1.

But she still knows exactly *how* it moves: a specific process (say, an $AR(1)$), with known parameters, forever.

She has swapped “I know the distribution” for “I know the distribution *of* the distribution.” She still knows everything. Just one level further up.

Four Fractures of the Invisible Hand



Visible fractures of the *invisible hand*

So Why Do Crashes Still Happen?

Merton's investor doesn't just know a distribution. She knows the distribution *of* the distribution — exactly how the opportunity set moves, forever.

If that's true, an event like any one of these four fractures should be rare, priced, and hedged away long before it happens.

Guttal, Raghavendra, Goel & Hoarau (2016) went looking for the specific statistical fingerprint of an approaching tipping point — **critical slowing down** - in all four. It's largely absent. Variability rises; the textbook warning sign does not.

If agents knew the distribution of the distribution, why should crashes like this happen at all - four times, a century apart, in nearly the same shape? That question is the door into everything that follows.

Frank Knight (1921) Draws a Line

Risk

You don't know which number comes up.

But you know it's a fair six-sided die.

Uncertainty

You don't know if it's a die.

Or how many sides it has.

Or whether someone is filing down a corner while you're not looking.

Mainstream finance — static *and* dynamic, Block 1 *and* Block 2 — lives entirely on the “risk” side of this line.

Keynes: “We Simply Do Not Know”

“We have, as a rule, only the vaguest idea of any but the most direct consequences of our acts.”

- Keynes, The General Theory, Ch. 12

Investors, on this view, don't compute expected utility, returns, over a known distribution.

They can only ascertain vaguely, influenced by *conventions*, and revise it “somewhat abruptly” when confidence shifts, not when a data point nudges a posterior.

The Beauty Contest

Keynes's newspaper competition: pick the six prettiest faces from a hundred photographs. The prize goes to whoever's picks match the *average* pick.

The winning strategy is not “pick who I find prettiest.”

It is: “pick who I think everyone else will think everyone else will pick.”

That is not a probability distribution over fundamentals. That is guessing about guessing, all the way down — and no amount of knowing the true distribution of anything helps you win.

Kalecki. The cost of borrowing doesn't just depend on the odds — it rises with how big your position already is, relative to your own capital. Risk is partly self-inflicted, by scale.

Minsky. Stability itself is what breeds instability. A long calm stretch quietly encourages everyone to take on more debt — which is exactly what makes the eventual reversal worse. The dynamics point the wrong way, on purpose.

even if mainstream *static* finance's checks its watch (like Samuelson or Merton)...

... you don't actually know what time zone you're in, whether it's been reset, or whether someone else's watch is about to change what time it is for *you*.

Part 5 The Whole Semester, in One Sentence

Two questions, asked of every idea, every week

The Course, Rebuilt From the Joke

Block	The clock problem	Weeks
1. Mainstream Static	No clock in the model at all.	1–4 Fisher → Tobin → Markowitz → CAPM
2. Mainstream Dynamic	A clock — and the agent knows exactly how it runs.	5–8 Euler equations, Hamiltonians, Samuelson, Merton
3. Heterodox	Nobody can know how the clock runs, or the clock is someone else's decision.	9–11 Keynes, Kalecki, Minsky, Keen, Brock–Hommes

Week 12: the three columns, set against each other, question by question.

1. Does this model have a genuine clock in it — a state, an expectation, a feedback — or just a subscript?
2. If yes, do the agent knows how the clock runs?

Ask both questions of every result, every week. By Week 12 you should be asking them before I do!

Franklin was right - he just left off the fine print!

“Time is money - but you have to actually put the time in to find out how much, and admit up front that you might not know the returns in advance.”

Let's start!

Week 1-4: Fisher's separation theorem and Tobin's two-fund separation model, an objective calculation, done first, with no calendar at all; the mean-variance frontier and the CAPM, closing out Block 1.

Before we end Block 1 - we will review the dynamics toolkit

From week 5: we start handing the models a working clock - both mainstream and heterodox versions

Questions?

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